

**Yr 9 Science
Physical Sciences
Test 1 - Waves**

NAME: SOLUTIONS

MARK: 25

1. Which of the following lists correctly arranges the different types of materials in order of increasing speed of sound (i.e. from slowest to fastest)?

- A Solids; liquids; gases
- B Solids; gases; liquids
- C Liquids; solids; gases
- D Gases; liquids; solids

2. Some types of electromagnetic radiation are said to be more 'energetic' than others. Which of the following options best describes what determines how 'energetic' electromagnetic radiation is said to be?

- A Higher frequency radiation is more 'energetic'.
- B Longer wavelength radiation is more 'energetic'.
- C Faster-moving radiation is more 'energetic'.
- D Radiation travelling in a vacuum is more 'energetic'.

3. Which of the following statements best describes the motion of particles in a transverse wave?

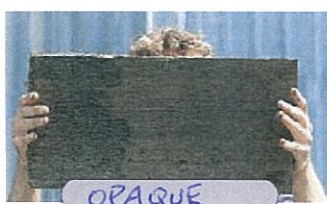
- A The particles vibrate parallel to the direction of energy transfer.
- B The particles vibrate at right angles to the direction of energy transfer.
- C The particles gradually spread out parallel to the direction of energy transfer.
- D The particles gradually spread out at right angles to the direction of energy transfer.

4. Which of the following options best describes a sound wave with a wavelength of 40 cm?

- A Compressions are 40 cm apart.
- B Compressions are 20 cm apart.
- C Compressions and rarefactions are 40 cm apart.
- D Compressions and rarefactions are 10 cm apart.

5.

Correctly label the diagrams
using the terms provided.



Opaque material

Translucent material

Transparent material

Reset

QUESTION	ANSWER
1	D
2	A
3	B
4	A
5	—
6	B
7	C
8	—
9	B
10	A
11	D

6. Which of the following options best describes the property of a sound that is determined by its amplitude?

- A Speed
- B Pitch
- C Loudness
- D Frequency

7. Which of the following options has the same meaning as a 'compression wave'?

- A A vibration wave.
- B An elastic wave.
- C A longitudinal wave.
- D A transverse wave.

8.

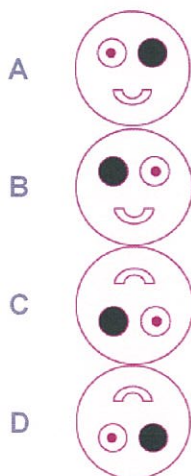
Label the diagram.

The diagram is divided into two horizontal sections. The top section, titled 'Water waves', shows a red sinusoidal wave on a blue background. A dashed horizontal line represents the 'Usual resting level of water'. A vertical double-headed arrow from the resting level to a peak is labeled 'AMPLITUDE'. A horizontal double-headed arrow between two consecutive peaks is labeled 'WAVELENGTH'. Below the wave, it says 'This is a TRANSVERSE wave'. The bottom section, titled 'Sound waves', shows a yellow background with black dots representing particle oscillations. A horizontal double-headed arrow between two regions of high dot density (compressions) is labeled 'Wavelength'. Below this, it says 'This is a COMPRESSION wave'. To the right of the diagram are four labeled buttons: 'Amplitude', 'Wavelength', 'compression', and 'transverse'. At the bottom right is a 'Reset' button.

9. Which of the following options best describes the property of a sound that is determined by its frequency?

- A Speed
- B Pitch
- C Loudness
- D Amplitude

10. A man has a black patch over his right eye. He looks into a plane mirror. Which of the following diagrams best describes the image he sees?



11. Which of the following explanations best describes the relation between a 'ray' and a 'beam'?

- A Rays travel in straight lines; beams are formed when rays are deflected.
B A beam is what is seen when rays are scattered.
C A beam shows the path of light; a ray is a bundle of beams.
D A ray shows the path of light; a beam is a bundle of rays.

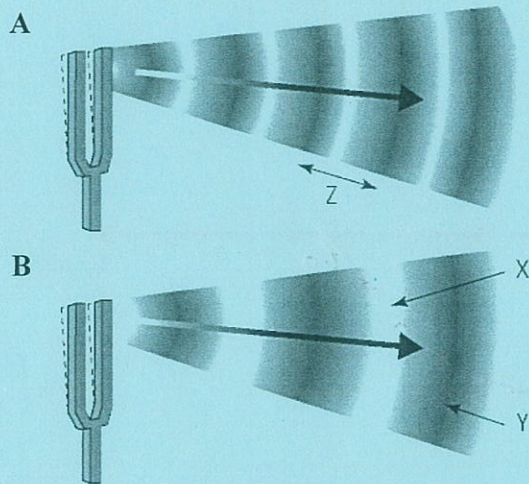
Energy Test

Student: SOLUTIONS Class: _____

Score ____ / 14

Wavelength and frequency

1. The following diagram shows two vibrating tuning forks producing sound waves.



- (a) Identify the features of the wave labelled X, Y and Z.

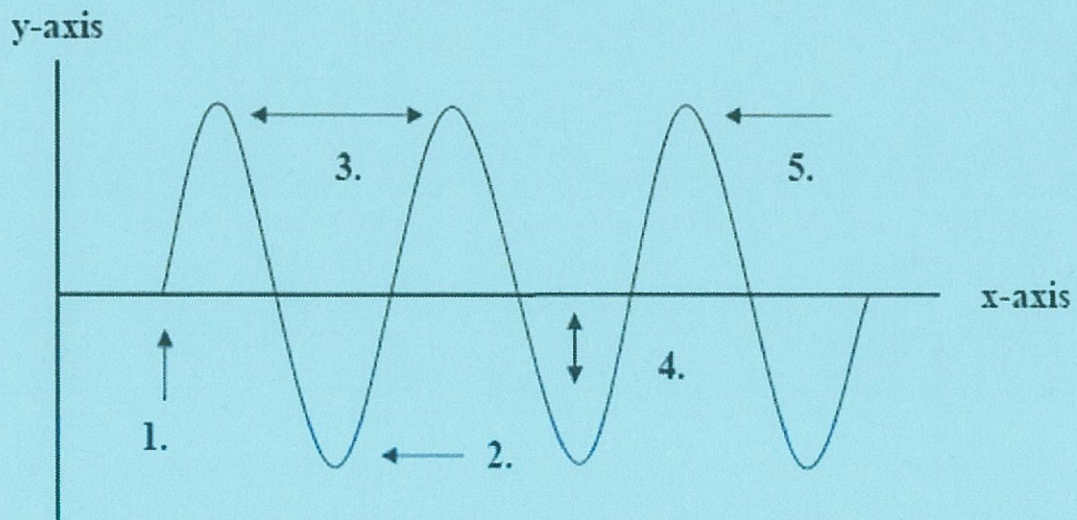
X: Rarefaction
 Y: Compression
 Z: Wavelength (3)

- (b) Which sound wave has the lowest pitch? Explain.

B
longer wavelength $\Rightarrow$ lower frequency. (2)

2. In the diagram below, identify the parts of a wave by using the provided definitions.

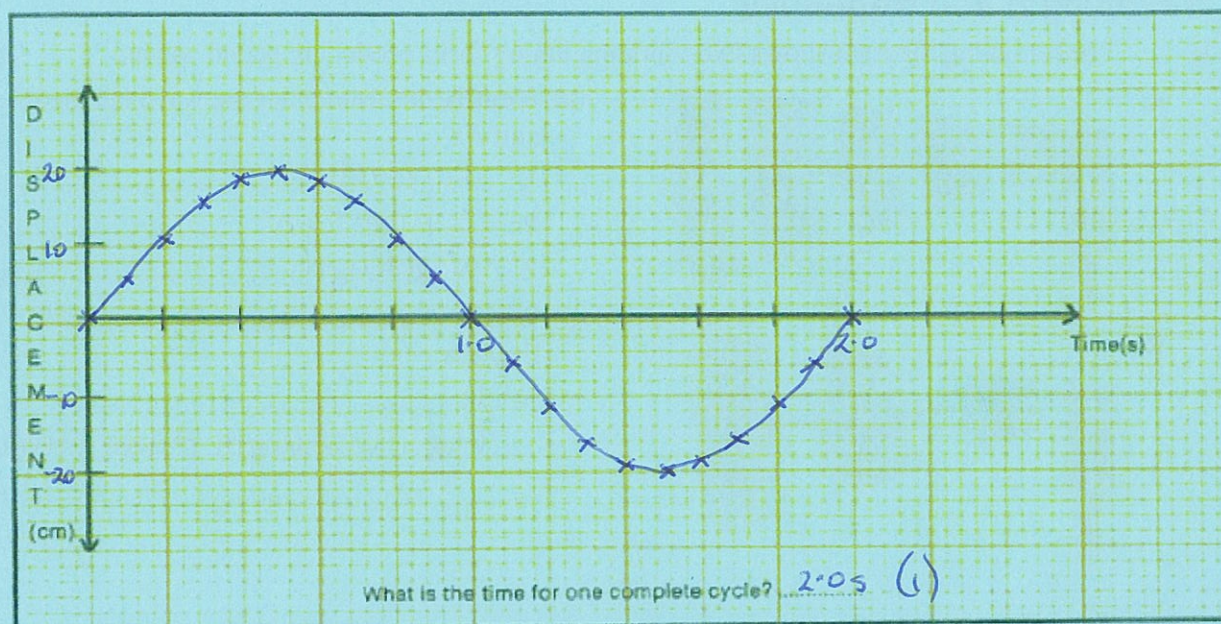
5 = **crest** The highest point of the wave above the line of origin.
 # 2 = **trough** The lowest point of the wave below the line of origin.
 # 1 = **line of origin** Signifies the original position of the medium.
 # 3 = **wavelength** The distance between two consecutive crests.
 # 4 = **amplitude** The distance from the line of origin to a crest or trough of a wave. (5)



The following data have been obtained by photo-analysis of the motion of a single particle in a medium as energy in a wave form passes through the medium.

Time (s)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
Displacement from rest position (cm)	0	5	11	16	19	20	19	16	11	5	0	-5	-11	-16	-19	-20	-19	-16	-11	-5	0

Plot the data on the graph paper.



Correct shape - 1 mark
 Correct plotting - 2 marks
 Correct time - 1 mark

(4)